

TEST REPORT

Applicant: Shenzhen Xunlong Software CO., Limited
Address: Room 219, Area 2, Block B, Shenzhen-Mingyou Purchasing Center,
Baoyuan Road, Xixiang Street, Bao'an, Shenzhen, Guangdong, China

The following sample(s) was/were submitted and identified on behalf of the client as:

Product name: Development Board
Model: OPI 5 Plus
Trade mark: Orange Pi™
Manufacturer: Orange Pi Electronic Technology (Dongguan) Co., Ltd
Address: Beike Base of Production, Education & Research, No.48, Changping
International Innovation Port, Huancheng North Road, Changping
Town, Dongguan City

Sample Received Date: 2024-03-12
Testing Period: 2024-03-12 ~ 2024-03-26

Test Requirement:

As specified by client, to determine the Lead(Pb), Cadmium(Cd), Mercury(Hg), Hexavalent Chromium (Cr(VI)), Polybrominated Biphenyls(PBBs), Polybrominated Diphenyl Ethers(PBDEs), Bis-(2-ethylhexyl) Phthalate (DEHP), Benzyl butyl Phthalate (BBP), Dibutyl Phthalate (DBP) and Diisobutyl Phthalate(DIBP) contents in the submitted sample(s) in accordance with RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU.

Test Conclusion:

Based on the performed tests on submitted sample(s), the test results of Lead(Pb), Cadmium(Cd), Mercury(Hg), Hexavalent Chromium(Cr(VI)), Polybrominated Biphenyls(PBBs), Polybrominated Diphenyl Ethers(PBDEs), Bis(2-ethylhexyl) phthalate (DEHP), Benzyl butyl Phthalate (BBP), Dibutyl Phthalate (DBP) and Diisobutyl Phthalate(DIBP) comply with the limits as set by RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU.

Test Result(s): Please refer to the following page(s);

Test Method: Please refer to the following page(s);

Compiled by: _____

Reviewed by: _____

Approved by: _____

Date: _____

2024-09-05



Sample Description:

No.	Part name	Description
1	PCBA	Blue PCB
2		Silvery metal shell of type-c interface
3		Black plastic of type-c interface
4		Metal plug pin of type-c interface
5		Silvery metal shell of network interface
6		Black adhesive cloth of network interface
7		Black plastic shell of network interface
8		Metal plug pin of network interface
9		Green diode light of network interface
10		Yellow diode light of network interface
11		Green PCB of network interface
12		SMD exclusion of network interface
13		SMD capacitor of network interface
14		Grey potting glue of network interface
15		Magnet core of network interface
16		Coil of network interface
17		White fiber of network interface
18		Core of wire of network interface
19		Silvery metal shell of HDMI interface
20		Black plastic of HDMI interface
21		Metal plug pin of HDMI interface
22		Silvery metal shell of USB20X2 interface
23		White plastic of USB20X2 interface
24		Metal contact pin of USB20X2 interface
25		Silvery metal shell of contact switch (small)
26		Black plastic button of contact switch (small)
27		Metal shrapnel of contact switch (small)
28		Colored black plastic of socket
29		Metal plug pin of socket
30		Beige plastic of beige interface
31		Metal plug pin of beige interface
32		Black plastic of black interface (large)
33		Metal plug pin of black interface (large)
34		Silvery metal shell of contact switch (small)
35		Black plastic button of contact switch (small)
36		Metal shrapnel of contact switch (small)
37		Silvery metal shell of USB30X2 interface
38		Blue plastic of USB30X2 interface
39		Metal contact pin of USB30X2 interface
40		Black body of audion

No.	Part name	Description
41	PCBA	Black rubber sleeve of MIC
42		Body of MIC
43		Black plastic shell of audio interface
44		Metal contact pin of audio interface
45		Yellow transparent adhesive tape of audio interface
46		Silvery metal stud of silvery metal stud
47		Silvery metal cover of CPU
48		PCB of CPU
49		SMD diode
50		SMD triode
51		SMD inductor
52		SMD crystal
53		SMD capacitor
54		SMD chip 1
55		SMD chip 2
56		SMD chip 3
57		Black plastic of black interface (small)
58		Metal plug pin of black interface (small)
59		Beige plastic of beige black interface
60		Black plastic of beige black interface
61		Metal plug pin of beige black interface
62A		White bar code label of label
63		Silvery metal shell of SD card slot
64		Black plastic of SD card slot
65		Metal plug pin of SD card slot

Test Result(s):
Lead(Pb), Cadmium(Cd), Mercury(Hg), Hexavalent Chromium (Cr(VI)), Polybrominated Biphenyls (PBBs), Polybrominated Diphenyl Ethers(PBDEs)

Part No.	Test Items	XRF Screening Result(mg/kg)	Chemical Test Result(mg/kg)	Conclusion
1	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
2	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	IN	N.D.	
	Br(PBBs&PBDEs)	/	/	
3	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
4	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
5	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
6	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	IN	N.D.	
7	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	IN	N.D.	

8	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
9	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
10	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
11	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	IN	N.D.	
12	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	IN	N.D.	
	Br(PBBs&PBDEs)	BL	/	
13	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
14	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
15	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	

16	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
17	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
18	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
19	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
20	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
21	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
22	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
23	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	IN	N.D.	

24	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
25	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	IN	N.D.	
	Br(PBBs&PBDEs)	/	/	
26	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
27	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	IN	N.D.	
	Br(PBBs&PBDEs)	/	/	
28	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	IN	N.D.	
29	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
30	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
31	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	

32	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
33	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
34	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
35	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
36	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	IN	N.D.	
	Br(PBBs&PBDEs)	/	/	
37	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
38	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	IN	N.D.	
39	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	

40	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	IN	N.D.	
41	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
42	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
43	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
44	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
45	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
46	Pb	OL	25963 ^{#1}	Pass
	Cd	IN	27	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
47	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	

48	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
49	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
50	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
51	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
52	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
53	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
54	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
55	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	

56	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
57	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
58	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
59	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
60	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
61	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	
62A	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
63	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	IN	N.D.	
	Br(PBBs&PBDEs)	/	/	

64	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	BL	/	
65	Pb	BL	/	Pass
	Cd	BL	/	
	Hg	BL	/	
	Cr(Cr(VI))	BL	/	
	Br(PBBs&PBDEs)	/	/	

Bis-(2-ethylhexyl) Phthalate (DEHP), Benzyl butyl Phthalate (BBP), Dibutyl Phthalate (DBP) and Diisobutyl Phthalate(DIBP)

Test Items	Result(mg/kg)		
	1+11+48	3+7+9+10	6
Bis-(2-ethylhexyl) Phthalate (DEHP)	N.D.	N.D.	N.D.
Benzyl butyl Phthalate (BBP)	N.D.	N.D.	N.D.
Dibutyl Phthalate (DBP)	N.D.	N.D.	N.D.
Diisobutyl Phthalate(DIBP)	N.D.	N.D.	N.D.
Conclusion	Pass	Pass	Pass

Test Items	Result(mg/kg)		
	14	17	20+23+26+28+30
Bis-(2-ethylhexyl) Phthalate (DEHP)	N.D.	N.D.	N.D.
Benzyl butyl Phthalate (BBP)	N.D.	N.D.	N.D.
Dibutyl Phthalate (DBP)	N.D.	N.D.	N.D.
Diisobutyl Phthalate(DIBP)	N.D.	N.D.	N.D.
Conclusion	Pass	Pass	Pass

Test Items	Result(mg/kg)		
	32+35+38+40+42	41	43+57+59+60
Bis-(2-ethylhexyl) Phthalate (DEHP)	N.D.	N.D.	N.D.
Benzyl butyl Phthalate (BBP)	N.D.	N.D.	N.D.
Dibutyl Phthalate (DBP)	N.D.	N.D.	N.D.
Diisobutyl Phthalate(DIBP)	N.D.	N.D.	N.D.
Conclusion	Pass	Pass	Pass

Test Items	Result(mg/kg)		
	45	62A	64
Bis-(2-ethylhexyl) Phthalate (DEHP)	N.D.	N.D.	N.D.
Benzyl butyl Phthalate (BBP)	N.D.	N.D.	N.D.
Dibutyl Phthalate (DBP)	N.D.	N.D.	N.D.
Diisobutyl Phthalate(DIBP)	N.D.	N.D.	N.D.
Conclusion	Pass	Pass	Pass

Note:

- 1.N.D. = Not Detected (<MDL)
MDL = Method Detection Limit
1mg/kg = 1ppm =0.0001%
/=Not Regulated or Not Applicable
2. BL = Below the XRF screening limit
IN = Further chemical test will be conducted when the screening result inconclusive
OL = Further chemical test will be conducted while the result is above the screening limit.
3. For metal samples, the sample is negative for Cr(VI), if the Cr(VI) concentration is less than 0.10 µg/cm², the coating is considered a non- Cr(VI) based coating;
The sample is positive for Cr(VI), if the Cr(VI) concentration is greater than 0.13 µg/cm²,
The sample coating is considered to contain Cr(VI);
The result is considered to be inconclusive, the Cr(VI) concentration is between the 0.10 µg/cm² and 0.13 µg/cm², unavoidable coating variations may influence the determination.
Because the storage condition and production date of the sample are not known, the test results of the sample of hexavalent chromium can only represent the state of hexavalent chromium in the samples tested.

Remark:

1. When conducting the test for PBBs&PBDEs, XRF was introduced to screen Br Exclusively; When conducting the test for Hexavalent Chromium, XRF was introduced to screen Chromium exclusively.
2. According to the client's statement , the material of the sample(s) comply with RoHS directive 2011/65/EU Annex III Exemption, Corresponding exemption clause:
#1 6(c) Lead is exempted as copper alloy containing up to 4% lead by weight .
3. Part No.62A Resubmitted Date: Mar. 22, 2024.

Test Method:

1. With reference to IEC 62321-1: 2013 Ed.1.0, IEC 62321-2:2021 Ed.2.0, IEC 62321-3-1:2013 Ed.1.0. XRF screening limits in mg/kg for regulated elements in various matrices.

Element	Limit of IEC 62321-3-1:2013 Ed.1.0 (mg/kg)		
	Polymers	Metals	Composite material
Pb	$BL \leq (700-3\sigma) < X$ $< (1300+3\sigma) \leq OL$	$BL \leq (700-3\sigma) < X$ $< (1300+3\sigma) \leq OL$	$BL \leq (500-3\sigma) < X$ $< (1500+3\sigma) \leq OL$
Cd	$BL \leq (70-3\sigma) < X$ $< (130+3\sigma) \leq OL$	$BL \leq (70-3\sigma) < X$ $< (130+3\sigma) \leq OL$	$LOD < X < (150+3\sigma)$ $\leq OL$
Hg	$BL \leq (700-3\sigma) < X$ $< (1300+3\sigma) \leq OL$	$BL \leq (700-3\sigma) < X$ $< (1300+3\sigma) \leq OL$	$BL \leq (500-3\sigma) < X$ $< (1500+3\sigma) \leq OL$
Cr	$BL \leq (700-3\sigma) < X$	$BL \leq (700-3\sigma) < X$	$BL \leq (500-3\sigma) < X$
Br	$BL \leq (300-3\sigma) < X$	/	$BL \leq (250-3\sigma) < X$

Note:

- BL= Below the XRF screening limit
- OL=Over the XRF screening limit
- X=The symbol "X" marks the region where further investigation is necessary.
- 3σ =The reproducibility of analytical instruments
- LOD= Detection limit

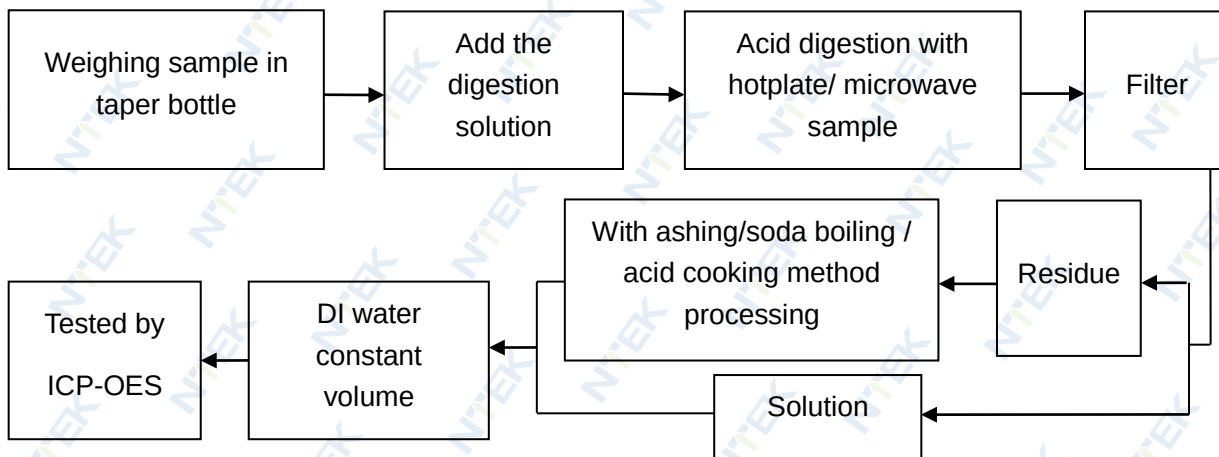
2. Chemical Test

Test item	Test method	Test instrument	MDL	Limit ^Δ
Lead (Pb)	IEC 62321-5:2013 Ed.1.0	ICP-OES	10 mg/kg	1000 mg/kg
Cadmium (Cd)	IEC 62321-5:2013 Ed.1.0	ICP-OES	10 mg/kg	100 mg/kg
Mercury (Hg)	IEC 62321-4:2013+AMD1:2017	ICP-OES	10 mg/kg	1000 mg/kg
Hexavalent Chromium(Cr(VI))	IEC 62321-7-1:2015 Ed.1.0	UV-Vis	0.10 µg/cm ²	1000 mg/kg
	IEC 62321-7-2:2017 Ed.1.0		8 mg/kg	
Polybrominated Biphenyls(PBBs)	IEC 62321-6:2015 Ed.1.0	GC-MS	100 mg/kg	1000 mg/kg
Polybrominated, Diphenyl Ethers(PBDEs)	IEC 62321-6:2015 Ed.1.0	GC-MS	100 mg/kg	1000 mg/kg
Bis-(2-ethylhexyl) Phthalate (DEHP)	IEC 62321-8:2017 Ed.1.0	GC-MS	50 mg/kg	1000 mg/kg
Benzyl butyl Phthalate (BBP)	IEC 62321-8:2017 Ed.1.0	GC-MS	50 mg/kg	1000 mg/kg
Dibutyl Phthalate (DBP)	IEC 62321-8:2017 Ed.1.0	GC-MS	50 mg/kg	1000 mg/kg
Diisobutyl Phthalate (DIBP)	IEC 62321-8:2017 Ed.1.0	GC-MS	50 mg/kg	1000 mg/kg

^ΔThe limit is quoted from RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU.

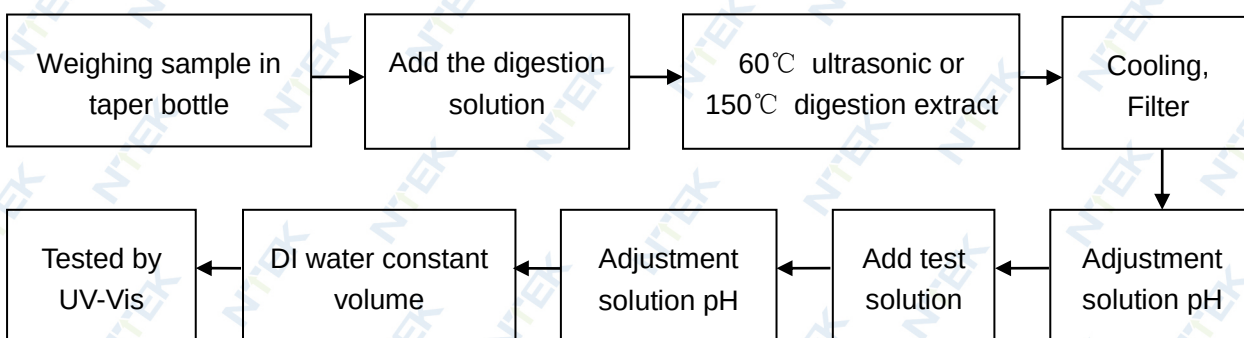
Test Flow:

1. Lead(Pb), Cadmium(Cd), Mercury (Hg)

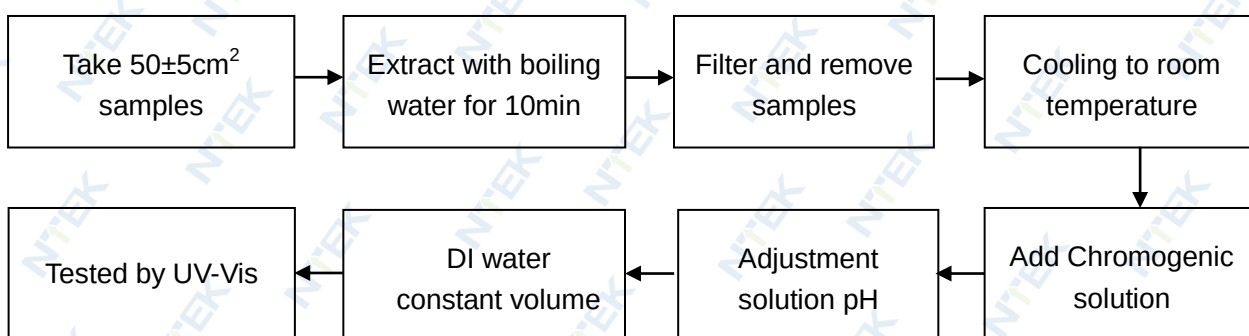


2. Hexavalent Chromium(Cr(VI))

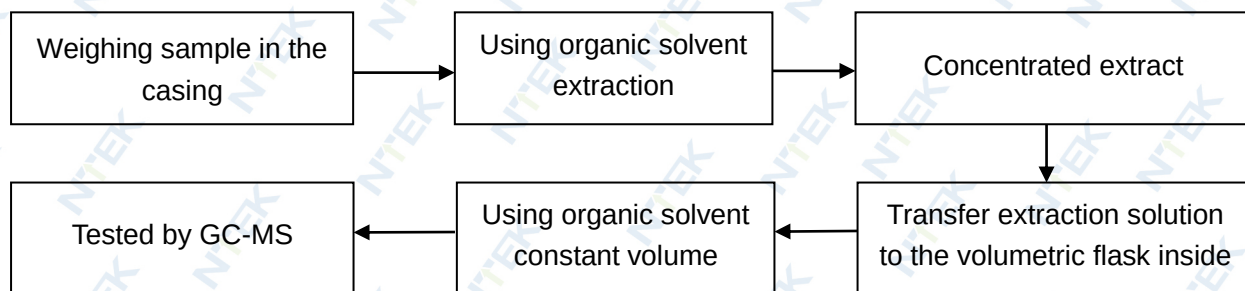
2.1 Non- metal sample(s)



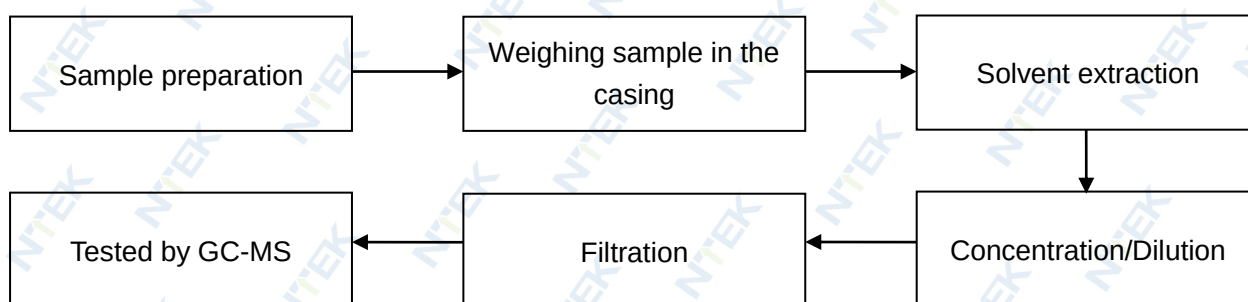
2.2 Metal sample(s)



3. PBBs/ PBDEs



4. Phthalates



Sample photo(s):



Fig.1



Fig.2

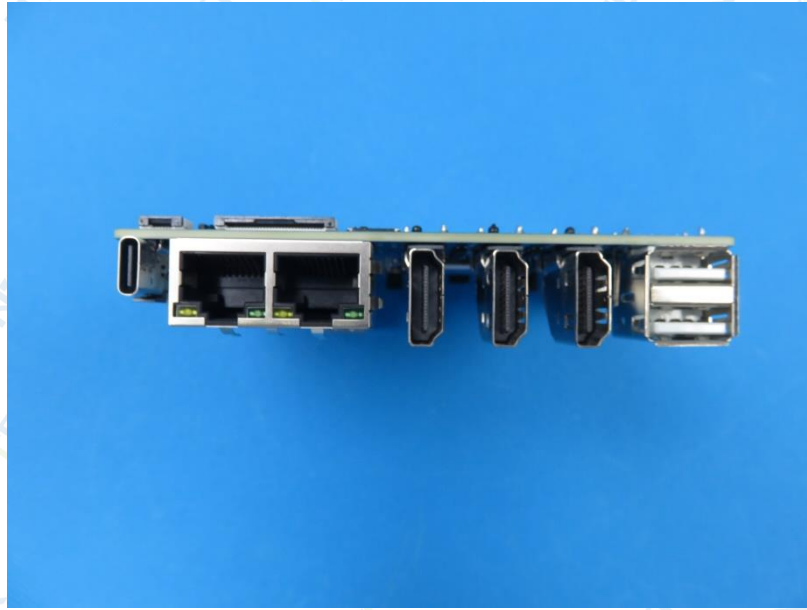


Fig.3

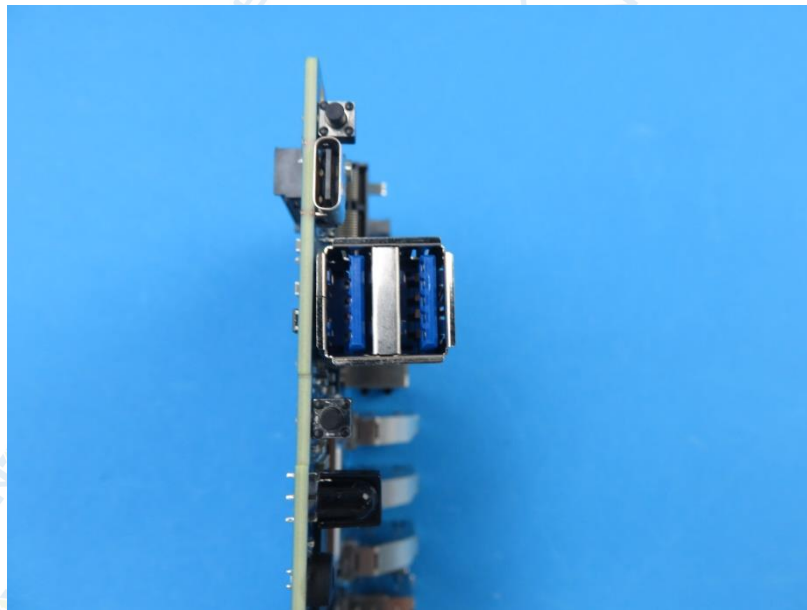


Fig.4

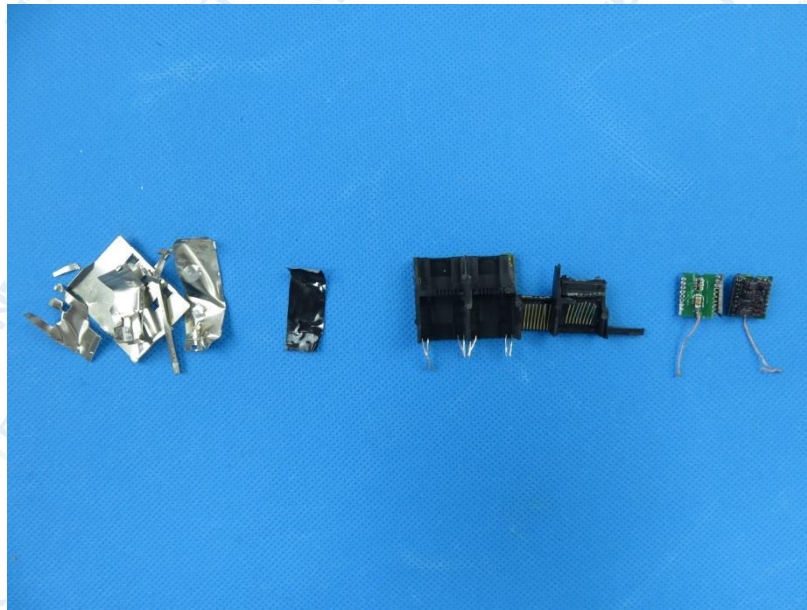


Fig.5

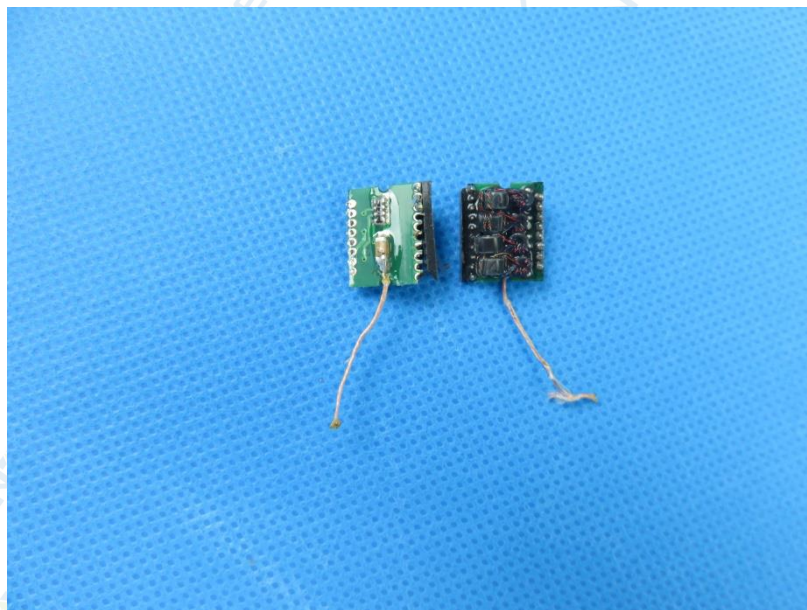


Fig.6

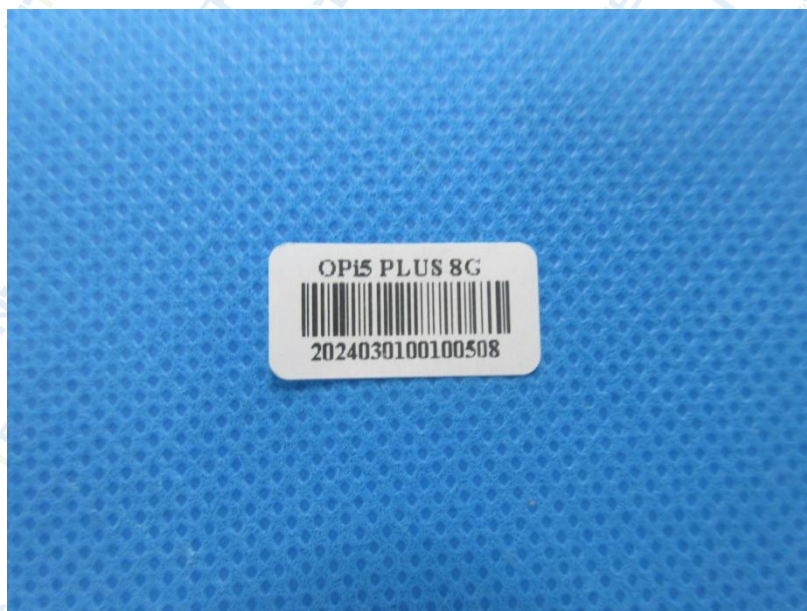


Fig.7

****End of Report****

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